CW 1

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```
library(tidyverse)
library(scales)
library(patchwork)
library(ggforce)
library(GGally)
library(knitr)

set.seed(7413)
pal_room <- c("Entire home/apt" = "#0072B2", "Private room" = "#E69F00")
```

1. Question and dataset context

What is the nightly price difference between entire homes and private rooms in the three London boroughs of Westminster, Hackney and Croydon?

Source This report uses the detailed London listings file published by Inside Airbnb, which collects publicly available Airbnb data and distributes it under a Creative Commons Attribution 4.0 licence.

Structure The dataset contains 91,954 listings, each row representing one individual Airbnb property, scraped between 18 and 23 September 2026.

Variables used This analysis uses each listing's borough, room type, advertised price, guest capacity, review score and availability.

Limitation	Explanation
Advertised, not actual price	The price column shows what the host asks for, not what a guest actually paid.
Self-selected hosts	Only people who chose to list on Airbnb appear here, so the data reflects Airbnb hosts, not London housing generally.
One snapshot in time	The data was collected over one week in September 2026, so it cannot show how prices change by season.
Approximate location	Airbnb shifts each listing's coordinates slightly for host privacy, so only borough-level location can be trusted.

Because of these limitations, the price comparisons in this report should be read as a description of what hosts in Westminster, Hackney and Croydon currently advertise, not as a measure of true market rents or of how much guests pay in practice.

2. Data preparation and workflow

```
listings <- read_csv("data/listings.csv", show_col_types = FALSE)

dim(listings)
```

```
## [1] 91954    90
```

```
#price column specifically:

listings |> select(price) |> head(10)
```

```
## # A tibble: 10 x 1
##    price
##    <chr>
##  1 $267.50
##  2 $127.00
##  3 $137.50
##  4 $228.00
##  5 <NA>
##  6 $290.00
##  7 $517.50
##  8 <NA>
##  9 <NA>
## 10 $59.00
```

```
class(listings$price)
```

```
## [1] "character"
```

Filtering the data and fixing price:

```r
boroughs <- c("Westminster", "Hackney", "Croydon")
room_types <- c("Entire home/apt", "Private room")

scope <- listings |>
  filter(neighbourhood_cleansed %in% boroughs,
         room_type %in% room_types) |>
  mutate(
    price = parse_number(price),     # Converting price col text to number
    borough = factor(neighbourhood_cleansed, levels = boroughs)
  )

dim(scope)
```

```
## [1] 18141    91
```

```r
#Checking for missing price values:

scope |> summarise(missing_price = mean(is.na(price)))
```

```
## # A tibble: 1 x 1
##   missing_price
##           <dbl>
## 1         0.342
```

```r
#checking why price is missing, which is the real insight:

scope |>
  mutate(no_availability = availability_365 == 0) |>
  group_by(no_availability) |>
  summarise(missing_price = mean(is.na(price)), n = n())
```

```
## # A tibble: 2 x 3
##   no_availability missing_price     n
##   <lgl>                   <dbl> <int>
## 1 FALSE                  0.0657 12651
## 2 TRUE                   0.978   5490
```

```r
grp <- scope |>
  group_by(borough, room_type) |>
  summarise(Listings = n(),
            `With price` = sum(!is.na(price)),
            `Missing price` = mean(is.na(price)),
            .groups = "drop")

grp |>
  mutate(`Missing price` = percent(`Missing price`)) |>
  rename(Borough = borough, `Room type` = room_type) |>
  kable(format.args = list(big.mark = ","), align = "llrrr",
        caption = "Table 1: Listings in scope, and how many lack a price.")
```

Table 2: Table 1: Listings in scope, and how many lack a price.

Borough	Room type	Listings	With price	Missing price
Westminster	Entire home/apt	8,783	6,637	24.4%
Westminster	Private room	1,951	1,262	35.3%
Hackney	Entire home/apt	3,793	1,962	48.3%
Hackney	Private room	1,937	858	55.7%
Croydon	Entire home/apt	909	656	27.8%
Croydon	Private room	768	567	26.2%

3. Exploratory visual analysis

3.1 Composition

```
shares <- scope |>
  count(borough, room_type) |>
  group_by(borough) |>
  mutate(share = n / sum(n)) |>
  ungroup()

ggplot(shares, aes(borough, share, fill = room_type)) +
  geom_col(position = "dodge") +
  scale_fill_manual(values = pal_room) +
  scale_y_continuous(labels = percent) +
  labs(x = NULL, y = "Share of listings", fill = "Room type")
```

Figure 1 shows that the mix of listing types varies sharply across the three boroughs: entire homes make up 82% of Westminster listings, 66% of Hackney's, but only 54% of Croydon's. A bar chart was chosen because position along a shared baseline is judged more accurately than the angles a pie chart would require (Cleveland and McGill, 1984). This imbalance matters for the main question: because Westminster is dominated by entire homes, any borough-level price comparison must account for room type separately rather than comparing boroughs as a whole.

3.2 Missing data

```
scope |>
  summarise(Price = mean(is.na(price)),
            `Review score` = mean(is.na(review_scores_rating)),
            Bedrooms = mean(is.na(bedrooms)),
            Accommodates = mean(is.na(accommodates))) |>
  pivot_longer(everything(), names_to = "variable", values_to = "share") |>
  mutate(variable = fct_reorder(variable, share)) |>
  ggplot(aes(share, variable)) +
  geom_point(size = 3) +
  scale_x_continuous(labels = percent) +
  labs(x = "Share of listings with a missing value", y = NULL)
```

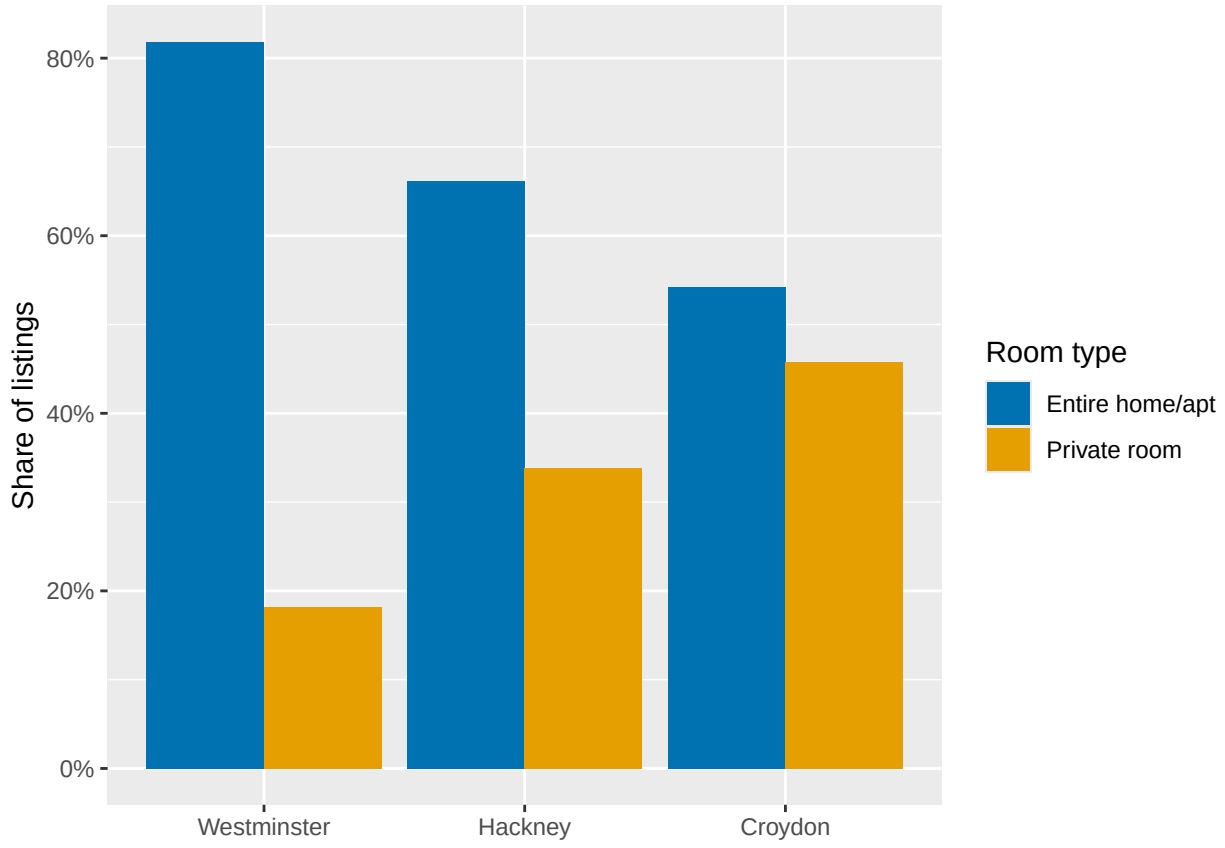


Figure 1: Figure 1: Share of entire homes vs private rooms, by borough.

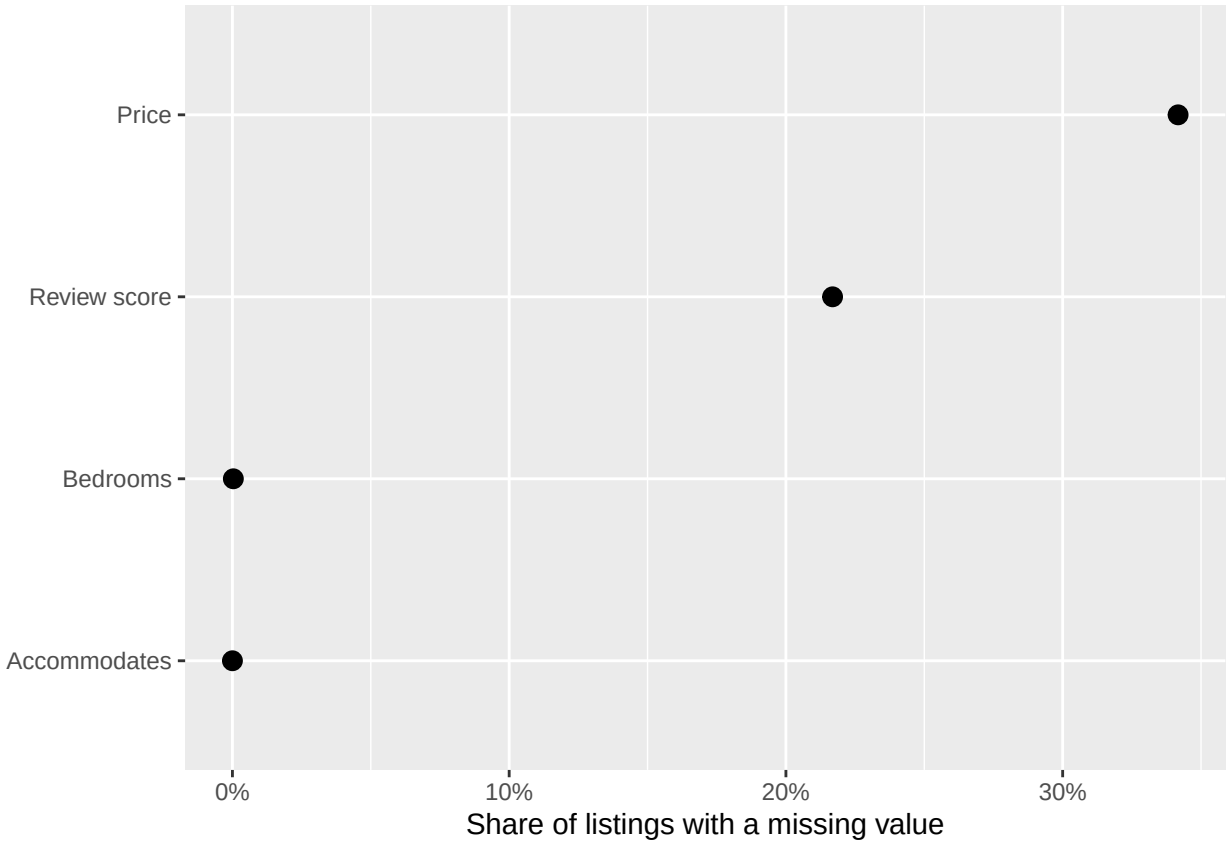


Figure 2: Figure 2: Share of missing values by variable.

Figure 2 ranks the four variables by how often they are missing. Price is the most incomplete at 34%, followed by review score at 22%; bedrooms and guest capacity are essentially complete. A dot plot was used rather than a bar chart because only the position of each point matters here, not an area — this keeps the chart simple and makes the ranking easy to read on one shared scale.

```
scope |>
  mutate(no_availability = availability_365 == 0) |>
  group_by(no_availability) |>
  summarise(share = mean(is.na(price)), .groups = "drop") |>
  mutate(no_availability = if_else(no_availability, "No availability", "Some availability")) |>
  ggplot(aes(no_availability, share)) +
  geom_col(width = .5) +
  scale_y_continuous(labels = percent) +
  labs(x = NULL, y = "Share with no price")
```

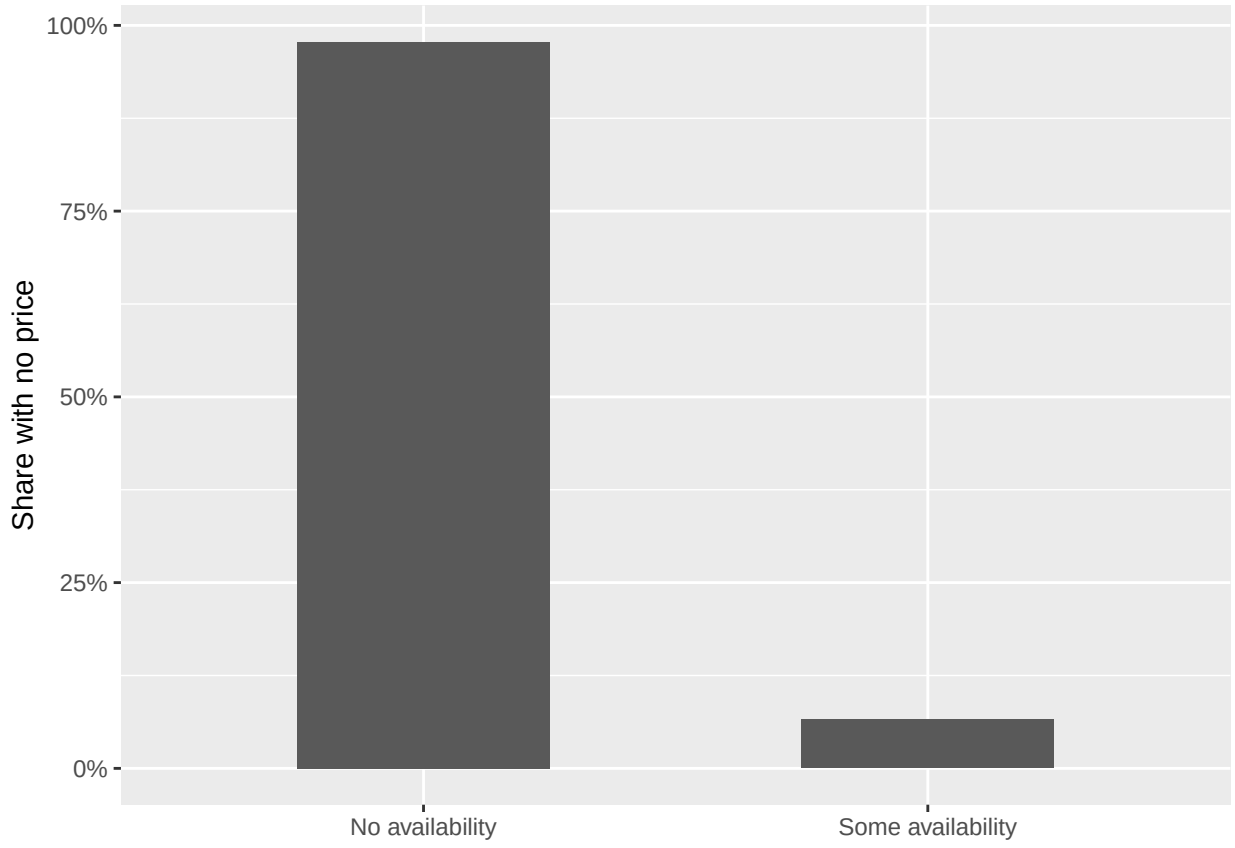


Figure 3: Figure 3: Share of listings with no price, by availability.

Figure 3 explains why price is missing so often: among listings with zero days available in the next year, 98% have no price, compared with just 7% of listings that have some availability. This indicates that Airbnb stops publishing a price once a listing is effectively inactive, rather than prices being missing at random. As a result, the comparisons in this report should be read as describing bookable listings, not the full stock of properties ever listed on Airbnb.

3.3 Distribution of price

```
dat <- scope |> filter(!is.na(price))

p_raw <- ggplot(dat, aes(price)) +
  geom_histogram(binwidth = 50, boundary = 0) +
  labs(title = "Raw scale", x = "Price per night", y = "Listings")

p_log <- ggplot(dat, aes(price)) +
  geom_histogram(bins = 40) +
  scale_x_log10() +
  labs(title = "Log scale", x = "Price per night", y = NULL)

p_raw + p_log
```

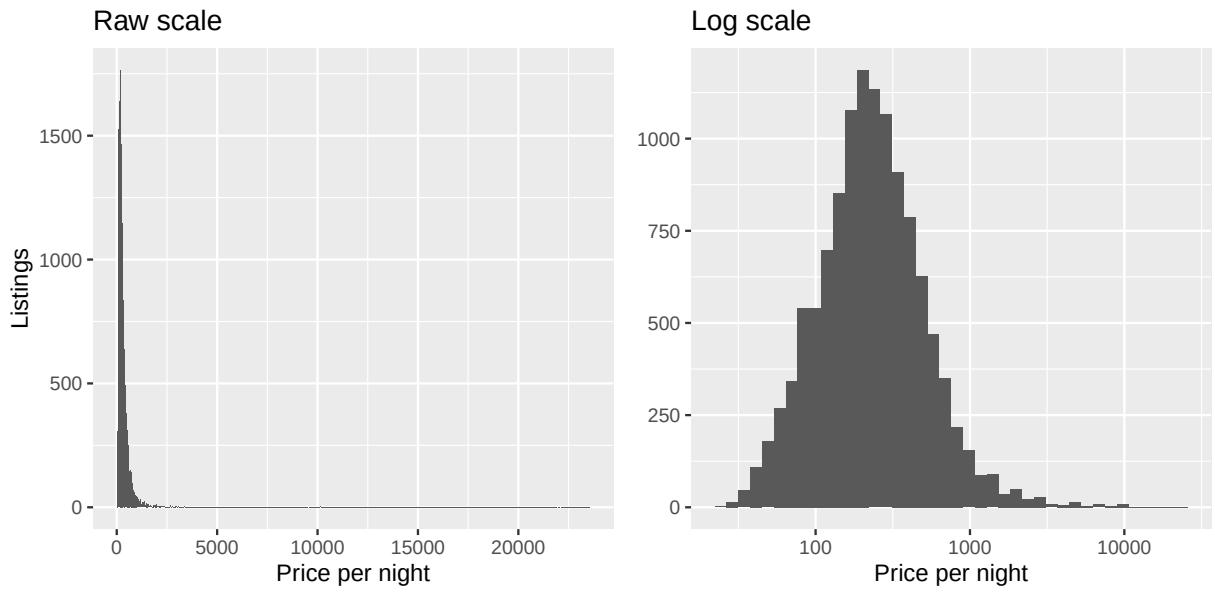


Figure 4: Figure 4: Nightly price on a raw scale (left) and log scale (right).

```
ggplot(dat, aes(price, fill = room_type, colour = room_type)) +
  geom_density(alpha = .35) +
  facet_wrap(~ borough) +
  scale_x_log10() +
  scale_fill_manual(values = pal_room) +
  scale_colour_manual(values = pal_room) +
  labs(x = "Price per night", y = "Density", fill = NULL, colour = NULL)
```

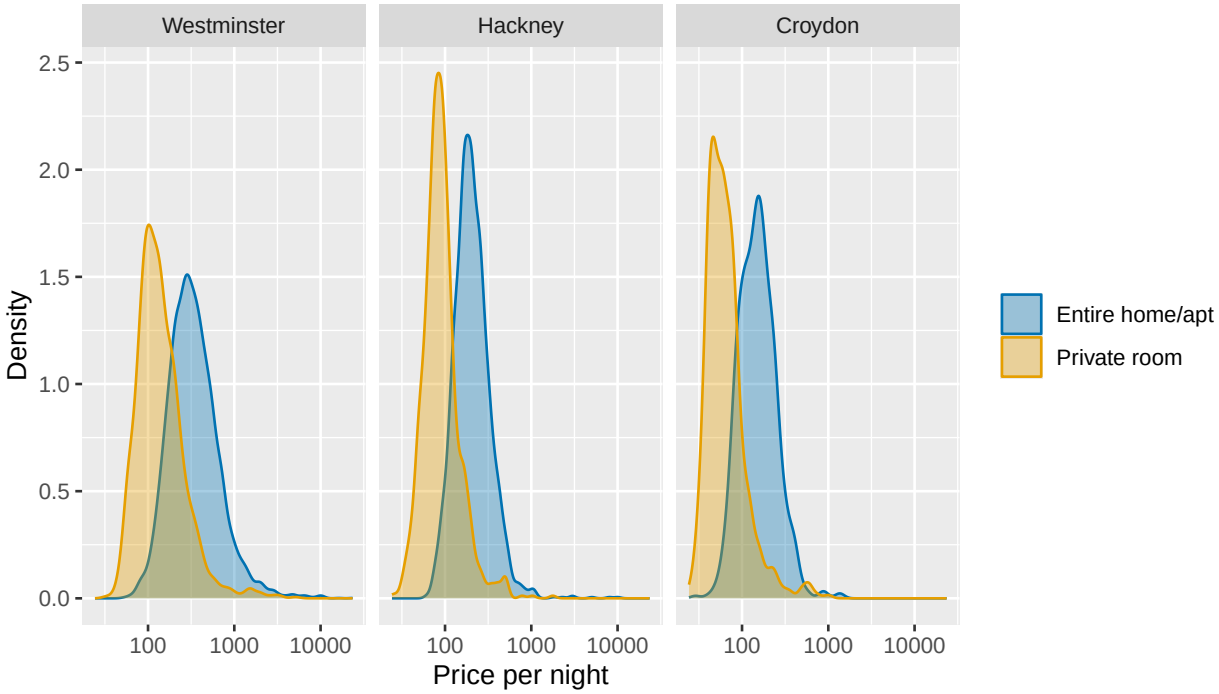


Figure 5: Figure 5: Density of price by room type, faceted by borough.